This assignment will not be graded and is only for practice.

Level 1:

1) Consider a square matrix $A = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & -1 \\ -1 & 0 & 1 \end{bmatrix}$. If

1 point

$P = -1M_{11} - 2M_{12} + 0M_{13}$
$Q = -2M_{21} + 0M_{22} + 1M_{23}$
$R = -1M_{31} - 0M_{32} + 1M_{33}$

where M_{ij} is the minor with respect to the (i, j)-th entry, then choose the set of correct options.

- [ ] $det(A) = -2.$
- [ ] $P = Q.$
- [ ] $P = Q = R.$
- [ ] $P = Q = R.$
- [ ] $Q = R$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$det(A) = -2.$
$P = Q = R.$

2) Suppose for a real 3×3 matrix A, there exists a real 3×3 matrix P such that $D = PAP^{-1}$ is a real 3×3 diagonal matrix. Choose the correct set of options.

1 point

- [ ] $det\ (A)$must be equal to $det\ (P)$.
- [ ] $det\ (A)$ must be equal to $det\ (D)$.
- [ ] The matrix A must be equal to D.
- [ ] If D is the identity matrix of order 3, then A must be the identity matrix of order 3

No, the answer is incorrect.
Score: 0

Accepted Answers:

$det\ (A)$ must be equal to $det\ (D)$.
If D is the identity matrix of order 3, then A must be the identity matrix of order 3

3) Choose the set of correct options for square matrices of order 3. ***1 point***

- [ ] If $AB = 0$, then one of the matrices between A and B must have determinant 0.
- [ ] If $AB = 0$, then both the matrices A and B must have determinant 0.
- [ ] If $AB = 3I$, where I denotes the identity matrix of order 3, then $detA = \frac{3}{det(B)}$.
- [ ] If $AB = 3I$, where I denotes the identity matrix of order 3, then $detA = \frac{9}{det(B)}$.
- [ ] If $AB = 3I$, where I denotes the identity matrix of order 3, then $detA = \frac{27}{det(B)}$.
- [ ] If $AB = 3I$, where I denotes the identity matrix of order 3, then both A and B must have non-zero determinant.

No, the answer is incorrect.
Score: 0

Accepted Answers:

If $AB = 0$, then one of the matrices between A and B must have determinant 0.
If $AB = 3I$, where I denotes the identity matrix of order 3, then $detA = \frac{27}{det(B)}$.
If $AB = 3I$, where I denotes the identity matrix of order 3, then both A and B must have non-zero determinant.

4) Consider a matrix $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$. If $a + b + c$ is divisible by 6 then choose the set of correct options. ***1 point***

- [ ] $det(A)$ is divisible by 5.
- [ ] $det(A)$ is divisible by 6.
- [ ] $det(A)$ is divisible by 2.

- [ ] $det(A)$ is divisible by 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$det(A)$ is divisible by 6.
$det(A)$ is divisible by 2.
$det(A)$ is divisible by 3.

5) Let $A = [a_{ij}]$ be a square matrix of order 3, where $a_{ij} = i$. Find $det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2:

6) Let $A = [\delta_{ij}]$ be a square matrix of order 3, where $\delta_{ij} = i \times j$. Find $det(A)$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

7) Consider the following square matrices:

1 point

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

Choose the set of correct options.

- [ ] $det(B) = det(A)$.
- [ ] $det(A) = -det(B)$.
- [ ] $det(C) = -2det(B)$.
- [ ] $det(C) = -2det(A)$.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$det(A) = -det(B)$.
$det(C) = -2det(B)$.
$det(C) = -2det(A)$.

(Based on the below information answer questions 8 and 9).

Let A be a real 3×3 matrix as given below:

$$A = [C_1 \quad C_2 \quad C_3]$$

where C_i represents the i-th column of the matrix A. Now let us consider the following set of matrices.

$$A_1 = [C_1 \quad C_2 + 5C_3 \quad C_3]$$
$$A_2 = [C_1 + C_2 + C_3 \quad C_2 \quad C_3]$$
$$A_3 = [C_1 \quad C_2 + 5C_3 \quad 0]$$
$$A_4 = [C_1 + C_2 \quad C_2 + C_3 \quad C_3 + C_1]$$

8) Which of the matrices have the same determinant as that of A? **1 point**

- [ ] A_1
- [ ] A_2
- [ ] A_3
- [ ] A_4

No, the answer is incorrect.
Score: 0

Accepted Answers:

A_1
A_2

9) Choose the set of correct options. **1 point**

- [ ] $det(A_3)$ cannot be determined from the given information.
- [ ] $det(A_3) = 0$
- [ ] $det(A_4) = 2\ det(A)$
- [ ] $det(A_2) = 3\ det(A)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$det(A_3) = 0$
$det(A_4) = 2\ det(A)$

10) If the diagonal entries of a 3×3 lower triangular matrix A are 1,2, and 3, then find the sum of roots of the equation $det(A - xI) = 0$, where I is the identity matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

Your score is: 0/10